

Branching:

1. Create a new branch **"GitNewBranch"**.
2. List all the local and remote branches available in the current trunk. Observe the "*" mark which denote the current pointing branch.

```
Nandu@Nandu MINGW64 ~/GitDemo (master)
$ git branch
  GitNewBranch
* master

Nandu@Nandu MINGW64 ~/GitDemo (master)
$ git branch GitNewBranch1

Nandu@Nandu MINGW64 ~/GitDemo (master)
$ git branch -a
  GitNewBranch
  GitNewBranch1
* master
remotes/origin/master

Nandu@Nandu MINGW64 ~/GitDemo (master)
```

3. Switch to the newly created branch. Add some files to it with some contents.

```
Nandu@Nandu MINGW64 ~/GitDemo (master)
$ git checkout GitNewBranch1
Switched to branch 'GitNewBranch1'

Nandu@Nandu MINGW64 ~/GitDemo (GitNewBranch1)
$ echo "Created in GitNewBranch1" > Branch.txt

Nandu@Nandu MINGW64 ~/GitDemo (GitNewBranch1)
$ git status
On branch GitNewBranch1
Untracked files:
  (use "git add <file>..." to include in what will be committed)
  Branch.txt

nothing added to commit but untracked files present (use "git add" to track)
```

4. Commit the changes to the branch.
5. Check the status with **"git status"** command.

```
Nandu@Nandu MINGW64 ~/GitDemo (GitNewBranch1)
$ git add Branch.txt
warning: in the working copy of 'Branch.txt', LF will
be replaced by CRLF the next time Git touches it

Nandu@Nandu MINGW64 ~/GitDemo (GitNewBranch1)
$ git commit -m "Added Branch.txt"
[GitNewBranch1 6c2420a] Added Branch.txt
1 file changed, 1 insertion(+)
create mode 100644 Branch.txt

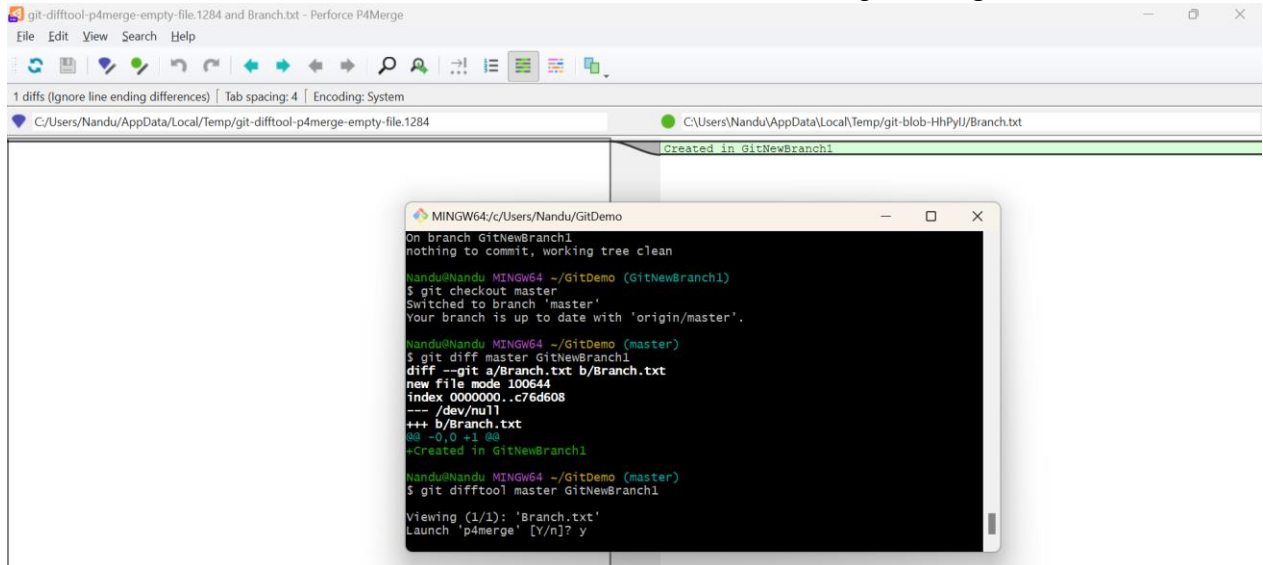
Nandu@Nandu MINGW64 ~/GitDemo (GitNewBranch1)
$ git status
On branch GitNewBranch1
nothing to commit, working tree clean
```

Merging:

1. Switch to the master.

```
Nandu@Nandu MINGW64 ~/GitDemo (GitNewBranch1)
$ git checkout master
Switched to branch 'master'
Your branch is up to date with 'origin/master'.
```

- List out all the differences between trunk and branch. These provide the differences in command line interface.
- List out all the visual differences between master and branch using **P4Merge** tool.



- Merge the source branch to the trunk.

```

Nandu@Nandu MINGW64 ~/GitDemo (master)
$ git merge GitNewBranch1
Updating 2385b75..6c2420a
Fast-forward
 Branch.txt | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 Branch.txt

```

- Observe the logging after merging using “**git log --oneline --graph --decorate**”

```

Nandu@Nandu MINGW64 ~/GitDemo (master)
$ git log --oneline --graph --decorate
* 6c2420a (HEAD -> master, GitNewBranch1) Added Branch.txt
* 2385b75 (origin/master) Removed Branch.txt
* c8b3d4f Added Branch.txt
* a90f877 Added gitignore file
* 1853b87 /*

```

- Delete the branch after merging with the trunk and observe the git status.

```

Nandu@Nandu MINGW64 ~/GitDemo (master)
$ git branch -d GitNewBranch1
Deleted branch GitNewBranch1 (was 6c2420a).

Nandu@Nandu MINGW64 ~/GitDemo (master)
$ git branch
GitNewBranch
* master

```

```

Nandu@Nandu MINGW64 ~/GitDemo (master)
$ git status
On branch master
Your branch is up to date with 'origin/master'.

nothing to commit, working tree clean

```